

The Katz Status Index

CLA365 TECHNICAL REPORT 2

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Abstract

The Katz Index of an adjacency matrix is the measure of the connectedness of the nodes, taking account of direct and indirect connections, where further connected nodes are dampened with attenuation factor α . This report discusses the Katz Status Index for social network analysis and implements it on a small scale office advice network to show the mechanisms at play, as well as on the top 990 Wikipedia articles to show its effectiveness considering large data sets. The index for each node is found by solving the Katz Matrix with the iterative Jacobi method. The Katz Status Index is a better measure of importance than degree centrality because the latter does not take into account the indirect connections of nodes that are two or more steps away.

1 Introduction

The Katz Status Index measures the importance of nodes in a graph by taking into account direct and indirect connections. In our method, we find the Katz Index for each node by solving a system of linear equations. This technique requires an adjacency matrix A which describes how nodes are connected in a graph. Then, the Katz matrix K is the infinite sum $K = \alpha A^T + \alpha^2 (A^T)^2 + \alpha^3 (A^T)^3 + \dots$ which takes into account indirect paths to each of the nodes with a weight based on the attenuation factor. This infinite sum is similar to a geometric sum which means it can be simplified to an exact answer, given that it converges. Therefore under the condition that $|\lambda| < 1$ for all eigenvalues of αA the infinite sum will converge.

The Katz status index $\vec{p}$ is found by $\vec{p} = K\vec{1}$. For the sake of brevity, K can be simplified as discussed such that we now have $\vec{p} = (I - \alpha A^T)^{-1} \alpha A^T \vec{1}$. Rearranging, we now have $(I - \alpha A^T) \vec{p} = \alpha A^T \vec{1}$, where $d = A^T \vec{1}$, the vector of in-degrees of the adjacency matrix. This can be further simplified by $B = (I - \alpha A^T)$, such that $B\vec{p} = \vec{d}$. Therefore, we have a system of linear equations that can be solved with a variety of methods.

In this report, we solve for the Katz status index $\vec{p}$ by using the iterative Jacobi method with an initial guess $\vec{p}_0 = \vec{1}$.

2 Example: Office Advice Network

To demonstrate Katz Centrality, consider the example where each person in an office is asked who they would turn to for advice. The graph of this situation will have an arrow drawn from node A to node B if person A would ask person B for advice. The adjacency matrix has an entry of 1 in position (i, j) if person i would ask person j for advice.

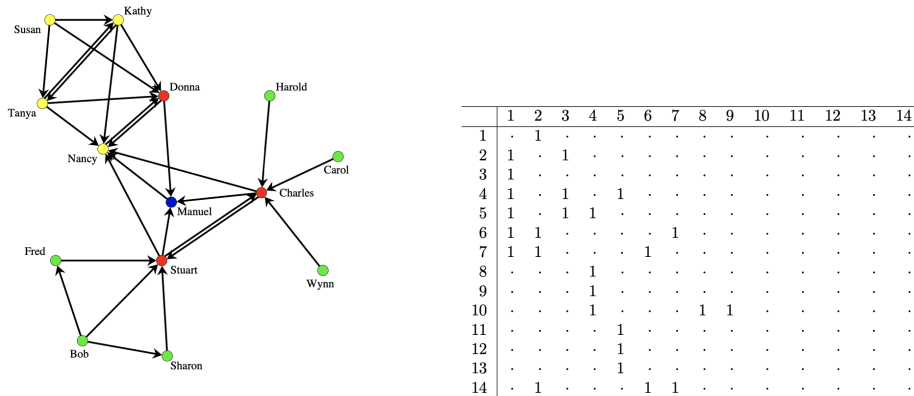


Figure 1: Office Advice Graph (left) and adjacency matrix A (right)

2.1 Selecting α

As stated before, for the sum represented by B to converge, it is necessary for the eigenvalues of αA to be strictly less than 1. For the adjacency matrix A in Figure 2, the largest value of α this is possible for is roughly 0.7548776. Thus, the sum represented by B converges if $\alpha \leq 0.7548776$. Higher values of α give greater weight to indirect connections in the graph whereas lower values give less weight to indirect connections. For example, in the graph in Figure 1, Susan is indirectly connected to Donna because Susan would ask Kathy for advice who would in turn ask Donna for advice. A higher attenuation factor would have a greater impact on the Katz Index for Donna's node because it puts greater weight on this indirect connection. The attenuation factor's impact on the Katz Index is interesting because for higher values of α , the relative importance of nodes is more concentrated among the nodes that represent people that a lot of others would go to for advice. On the flip side, low values of α produce a relatively more even distribution of the Katz Index among the nodes. Figure 2 shows the Jacobi iterations for $\alpha = 0.1$. Figures 3 and 4 show the Katz Index of each person using varying values of α .

Names	X0	X1	X2	X3	X4	X5	X6	X7	X8	X9
Nancy	1	1.200	0.958	0.846	0.825	0.822	0.822	0.822	0.822	0.822
Donna	1	0.600	0.500	0.444	0.429	0.427	0.427	0.427	0.427	0.427
Manuel	1	0.600	0.518	0.447	0.436	0.434	0.434	0.434	0.434	0.434
Stuart	1	0.778	0.489	0.467	0.467	0.467	0.467	0.467	0.467	0.467
Charles	1	0.800	0.478	0.449	0.447	0.447	0.447	0.447	0.447	0.447
Kathy	1	0.400	0.240	0.224	0.222	0.222	0.222	0.222	0.222	0.222
Tanya	1	0.400	0.240	0.224	0.222	0.222	0.222	0.222	0.222	0.222
Fred	1	0.200	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100
Sharon	1	0.200	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100
Bob	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carol	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Harold	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Wynn	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Figure 2: Jacobi iterations for $\alpha = 0.1$

Names	X0.01	X0.1	X0.2	X0.3	X0.4	X0.5	X0.6	X0.7	X0.7548776
Nancy	0.062	0.822	2.333	5.213	11.109	24.8	65.445	307.006	3025.485
Donna	0.031	0.427	1.267	2.978	6.710	15.9	44.667	223.528	2272.355
Manuel	0.031	0.434	1.277	2.922	6.316	14.2	37.528	175.517	1724.949
Stuart	0.041	0.467	1.100	1.971	3.200	5.0	7.800	12.600	16.968
Charles	0.040	0.447	1.020	1.791	2.880	4.5	7.080	11.620	15.828
Kathy	0.020	0.222	0.500	0.857	1.333	2.0	3.000	4.667	6.159
Tanya	0.020	0.222	0.500	0.857	1.333	2.0	3.000	4.667	6.159
Fred	0.010	0.100	0.200	0.300	0.400	0.5	0.600	0.700	0.755
Sharon	0.010	0.100	0.200	0.300	0.400	0.5	0.600	0.700	0.755
Bob	0.000	0.000	0.000	0.000	0.000	0.0	0.000	0.000	0.000
Carol	0.000	0.000	0.000	0.000	0.000	0.0	0.000	0.000	0.000
Harold	0.000	0.000	0.000	0.000	0.000	0.0	0.000	0.000	0.000
Wynn	0.000	0.000	0.000	0.000	0.000	0.0	0.000	0.000	0.000
Susan	0.000	0.000	0.000	0.000	0.000	0.0	0.000	0.000	0.000

Figure 3: Unscaled Katz Status Index for various values of α

Names	X0.01	X0.1	X0.2	X0.3	X0.4	X0.5	X0.6	X0.7	X0.7548776
Nancy	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Donna	0.502	0.519	0.543	0.571	0.604	0.641	0.683	0.728	0.751
Manuel	0.503	0.528	0.548	0.561	0.569	0.573	0.573	0.572	0.570
Stuart	0.657	0.568	0.472	0.378	0.288	0.202	0.119	0.041	0.006
Charles	0.653	0.543	0.437	0.344	0.259	0.181	0.108	0.038	0.005
Kathy	0.327	0.270	0.214	0.164	0.120	0.081	0.046	0.015	0.002
Tanya	0.327	0.270	0.214	0.164	0.120	0.081	0.046	0.015	0.002
Fred	0.162	0.122	0.086	0.058	0.036	0.020	0.009	0.002	0.000
Sharon	0.162	0.122	0.086	0.058	0.036	0.020	0.009	0.002	0.000
Bob	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carol	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Harold	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Wynn	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Susan	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Figure 4: Scaled Katz Status Index for various values of α

2.2 Katz Status Index and In-Degree Rank

The Katz Status Index and the in-degree rank are related, though they are not always the same. For the office advice network, the in-degree rank is $\vec{d} = (6, 3, 3, 4, 4, 2, 2, 1, 1, 0, 0, 0, 0, 0)$. Nancy, the first entry into the vector of the Katz Status Index and the in-degree rank, always has the highest relative importance, and Bob, Carol, Harold, Wynn, and Susan have no relative importance in either system. For small attenuation values, Stuart and Charles are still the 2nd and 3rd most sought after people according to the Katz Status Index as they are using in-degree rank (although they have the same value in the in-degree rank and different in the Katz Status Index). As the attenuation values get higher, Donna and Manuel become the 2nd and 3rd most sought after people according to the Katz Status Index whereas they are 4th and 5th in the in-degree rank. Based on these results, it is apparent that higher weighting of indirect connections does have an impact on the relative importance of a node.

2.3 Derivation of Jacobi iterations

The Jacobi method is an iterative method for solving a system of linear equations. The user provides an initial guess and subsequent iterations approach the correct solution. Our choice of α provides that the Jacobi method will converge.

To solve using the Jacobi method, the matrix A is separated into D , a matrix containing just the diagonal of A , and $R = A - D$, the remainder.

The system of equations we are solving is $B\vec{p} = \vec{d}$, where $B = \alpha^{-1}I - A^T$. Thus, we know D and R for the Jacobi Method are $D = \alpha^{-1}I$ and $R = -A^T$. This is due to our adjacency matrix not including self-connections, meaning the diagonal of A is empty.

The Jacobi iteration following $\vec{p}_k$ is calculated as follows: $\vec{p}_{k+1} = D^{-1}(\vec{d} - R\vec{p}_k)$. Plugging in our result for D above, we get $\vec{p}_{k+1} = \alpha I(\vec{d} + A^T\vec{p}_k) = \alpha\vec{d} + \alpha A^T\vec{p}_k$. Taking into account the fact that $\vec{d} = A^T\vec{1}$, this becomes $\vec{p}_{k+1} = \alpha A^T(\vec{1} + \vec{p}_k)$. To illustrate this, consider taking $k = 0$. For this problem we have $\vec{d} = (6, 3, 3, 4, 4, 2, 2, 1, 1, 0, 0, 0, 0, 0)$ and a starting guess $\vec{p}_0 = \vec{1}$, with $\alpha = 0.1$. The result is the same vector as in iteration 1 of Figure 2:

$$\vec{p}_{k+1} = \alpha A^T(\vec{1} + \vec{1}) = \alpha A^T\vec{2} = 2\alpha A^T\vec{1} = 0.2\vec{d} = (1.2, 0.6, 0.6, 0.8, 0.8, 0.4, 0.4, 0.2, 0.2, 0, 0, 0, 0, 0).$$

3 Wikipedia 990

The Katz Status Index can be used on larger scales to provide more robust social network analysis. In this section, we will use the same principles discussed in the office advice network example and apply it to the top 990 Wikipedia articles to ascertain which one is the most connected. The data set includes only the top 990 sets, and contained information on which sites had links to other sites on the list. The Katz Status Index was then computed for various attenuation factors.

3.1 Selecting α

As stated before, the magnitude of the eigenvalues of αA need to be strictly less than one for the matrix-solve to converge.

In order to find the threshold α -value we created a function which allowed various inputs of α , outputting a boolean value indicating convergence for a particular α . The R-function is shown below.

```

1 getConverges = function(A, alpha) {
2   eig = eigen(alpha * A)$values
3   return (all(abs(eig) < 1))
4 }

```

Therefore, we are able to automatically test many values of α . The eigenvalue calculation is slow, so we tested 20 values at a time, narrowing the search with each result. Therefore, we find that for $\alpha \leq 0.00239$ there is convergence.

The maximum attenuation factor is much smaller in this instance, when compared to the office advice example, because there are many more nodes and thus many more indirect connections.

3.2 Analysis

The Katz Status Index is calculated for varying convergent values of α , and the ten highest values in $\vec{p}$ are shown for the computations with varying attenuation factors (one relatively small, one relatively medium, and one relatively large) in Figure 6. As was the case in the Office Advice Network example, lower attenuation factors have a relatively more even distribution of importance as measured by the Katz Status Index, whereas higher values of α concentrated the relative importance amongst the top few.

The article for the year 2007 is consistently the most Katz-connected article regardless of the attenuation factor. The other Wikipedia articles in the top ten change based on the relative strength of the attenuation factor, dependent on whether there were more indirect or direct hyperlink references. An interesting phenomenon is the frequent occurrence of articles for dates in the top ten, especially for higher values of α . Many articles reference dates, such as a specific day in the year or a year in general. Therefore, there are many direct and indirect connections to these articles, explaining the shown results. Other articles that are in the top ten for various attenuation factors are the United States, United Kingdom, and World War II.

Site	Rating	Site	Rating	Site	Rating
2007	1.0000000	2007	1.0000000	2007	1.0000000
United_States	0.9586411	United_States	0.9646534	March_4	0.9766742
2006	0.8951258	2006	0.9254573	United_States	0.9758466
2008	0.8463810	2005	0.8428754	April_1	0.9748662
United_Kingdom	0.7799114	2008	0.8295213	2006	0.9745121
2005	0.7769576	January_1	0.7999806	January_1	0.9739841
World_War_II	0.6942396	October_1	0.7939767	December_31	0.9731234
January_1	0.6927629	December_31	0.7906017	March_17	0.9729612
2004	0.6912856	July_1	0.7873702	May_19	0.9717846
October_1	0.6839003	May_19	0.7872349	March_1	0.9711212

Figure 5: Top 10 Wikipedia Sites by Katz Status Index for attenuation values from left to right of $\alpha = 0.00000001$, $\alpha = 0.00115$, and $\alpha = 0.0023$.

4 Conclusion

The Katz Status Index is an effective way to measure the relative importance of different nodes in a social network by taking into account indirect connections. Manipulating the attenuation factor impacts the distribution of the relative importance of nodes; higher values of α result in all nodes being more connected and receiving increased status indices—likewise, lower values lead to lower connection, with relatively fewer nodes having as much ‘importance’. Based on the analysis of the top 990 Wikipedia sites, the most important website according to the Katz Status Index is 2007 regardless of choice of α . This occurs because dates are connected to many different Wikipedia articles directly and indirectly.

References

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